

Newton–Schulz Dynamics Beyond Norm Bounds: Full Spectral Basins, Jordan Rates, and Moore–Penrose Compatibility

HCS-C317 theorem-and-evidence package

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Abstract

For $X_{k+1} = X_k(2I - AX_k)$ we classify exact-arithmetic convergence without replacing spectral radius by a sufficient norm bound. In the invertible case, the basin is exactly $\rho(I - AX_0) < 1$, with a sharp Jordan rate and a complete unit-circle boundary. For arbitrary rectangular A , convergence to $A^\dagger$ holds if and only if the initial matrix has Moore–Penrose support and its range-compressed residual lies in the open spectral disk. The canonical start αA^* is resolved on the open corridor, both endpoints, every exterior face, repeated maximal singular values, and rank zero. Finite exact matrices audit the algebra but do not replace the arbitrary-dimensional proof.

1 Exact residual dynamics

Fix $A \in \mathbb{C}^{m \times n}$ and $X_0 \in \mathbb{C}^{n \times m}$, and define

$$X_{k+1} = X_k(2I_m - AX_k). \quad (1)$$

This reciprocal-matrix iteration goes back to Schulz [1] and Hotelling [2]; generalized-inverse iterations were studied by Ben-Israel [3].

Theorem 1 (Full square basin). *Suppose $m = n$ and A is invertible. With $R_0 = I - AX_0$,*

$$R_k = R_0^{2^k}, \quad X_k = A^{-1}(I - R_0^{2^k}). \quad (2)$$

Thus $X_k \rightarrow A^{-1}$ if and only if $\rho(R_0) < 1$. Whenever $\rho = \rho(R_0) > 0$, let s_ be the largest Jordan-block size among eigenvalues of modulus ρ . Then, for every matrix norm,*

$$\|R_k\| = \Theta\left((2^k)^{s_*-1} \rho^{2^k}\right). \quad (3)$$

For $\rho = 0$ there is finite termination once 2^k reaches the nilpotency index. For $\rho = 1$, semisimple peripheral blocks give a bounded, nonvanishing residual, whereas a defective peripheral block of maximal size $s_ > 1$ gives $\Theta((2^k)^{s_*-1})$ growth. For $\rho > 1$, (3) is polynomial-times-double-exponential divergence.*

Proof. Multiplying (1) by A gives $I - AX_{k+1} = (I - AX_k)^2$, proving (2). Matrix powers tend to zero exactly when their spectral radius is below one. For a Jordan block $J = \lambda I + N$ of size s and an integer $\ell \geq 0$,

$$J^\ell = \sum_{j=0}^{\min(s-1, \ell)} \binom{\ell}{j} \lambda^{\ell-j} N^j. \quad (4)$$

At $\ell = 2^k$, the largest nonzero peripheral term has order $\ell^{s_*-1} \rho^\ell$; equivalence of finite-dimensional norms transfers the two-sided estimate through a fixed similarity. Formula (4) also gives every boundary statement. In particular, “bounded and nonvanishing” at a semisimple unit-circle block does not assert that the residual sequence itself fails to converge. $\square$

2 The arbitrary-rank Moore–Penrose basin

Let $A^\dagger$ be the Moore–Penrose inverse and $P = AA^\dagger$, $Q = A^\dagger A$.

Theorem 2 (Compatibility is necessary and sufficient). *The iterates in (1) converge to $A^\dagger$ if and only if*

$$X_0 = QX_0P, \quad \rho((P - AX_0)|_{\text{Ran } A}) < 1. \quad (5)$$

Under (5), with $R_0 = P - AX_0$,

$$R_k = R_0^{2^k}, \quad X_k = A^\dagger - A^\dagger R_0^{2^k}. \quad (6)$$

Proof. Compatibility is invariant. It implies $PR_kP = R_k$ and $A^\dagger AX_k = X_k$, so the same calculation as before gives $R_{k+1} = R_k^2$ and $X_k = A^\dagger - A^\dagger R_k$. This proves sufficiency.

For necessity, use unitary SVD coordinates and write

$$A = \begin{pmatrix} \Sigma & 0 \\ 0 & 0 \end{pmatrix}, \quad X_k = \begin{pmatrix} B_k & C_k \\ D_k & E_k \end{pmatrix},$$

where Σ is positive diagonal. Direct multiplication gives

$$B_{k+1} = B_k(2I - \Sigma B_k), \quad C_{k+1} = (2I - B_k \Sigma)C_k, \quad (7)$$

$$D_{k+1} = D_k(2I - \Sigma B_k), \quad E_{k+1} = 2E_k - D_k \Sigma C_k. \quad (8)$$

If $X_k \rightarrow A^\dagger$, then $B_k \rightarrow \Sigma^{-1}$ and $(I - \Sigma B_0)^{2^k} \rightarrow 0$, proving the spectral condition. Moreover $2I - B_k \Sigma = I + \Sigma^{-1}(I - \Sigma B_k) \Sigma$ and $2I - \Sigma B_k = I + (I - \Sigma B_k)$. Their deviations from I are summable because the residual powers decay geometrically after a finite index; every factor is invertible because its shifted residual has spectrum strictly inside the unit disk. The associated left and right infinite products therefore converge to invertible matrices. Equations (7)–(8) can tend to the required zero off-support blocks only when $C_0 = D_0 = 0$. Then $E_k = 2^k E_0$, hence $E_0 = 0$. These three equalities are precisely $X_0 = QX_0P$. $\square$

3 The canonical scalar corridor

Theorem 3 (All αA^* faces). *Let $A \neq 0$, let $\sigma_{\max}$ be its largest singular value, and take real α . For $X_0 = \alpha A^*$,*

$$X_k \longrightarrow A^\dagger \iff 0 < \alpha < \frac{2}{\sigma_{\max}^2}.$$

At equality, every maximal-singular-value direction is deleted after one step and remains zero, while each smaller direction whose residual has modulus below one tends to its inverse coefficient. At $\alpha = 0$, zero is fixed. If $\alpha < 0$ or $\alpha > 2/\sigma_{\max}^2$, a maximal direction diverges. If $A = 0$, the canonical start is zero for all α and converges trivially; among arbitrary starts, only $X_0 = 0$ converges.

Proof. On a nonzero singular direction σ , the initial residual is $r_0 = 1 - \alpha\sigma^2$ and the iterate coefficient is

$$x_k = \frac{1 - r_0^{2^k}}{\sigma}.$$

All directions converge to $1/\sigma$ exactly in the open corridor. At the sharp upper endpoint, $r_0 = -1$ on every maximal direction, so $x_1 = 0$ and thereafter $r_k = 1$. The remaining claims follow directly; when $A = 0$, (1) reduces to $X_{k+1} = 2X_k$. $\square$

Revision certificate: Moore–Penrose iff basin and alpha faces

This round adds the necessity of support compatibility and every canonical scalar boundary, including rank zero.

References

- [1] G. Schulz, “Iterative Berechnung der reziproken Matrix,” *ZAMM* 13 (1933), DOI: 10.1002/zamm.19330130111.
- [2] H. Hotelling, “Some new methods in matrix calculation,” *Ann. Math. Stat.* 14 (1943), DOI: 10.1214/aoms/1177731489.
- [3] A. Ben-Israel, “An iterative method for computing the generalized inverse of an arbitrary matrix,” *Math. Comp.* 19 (1965), DOI: 10.1090/S0025-5718-1965-0179915-5.